New England Distributed Energy Resources Optimization Network

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EC417: Electric Energy Systems

Context of the Problem

The rapid growth of data centers is placing increasing demand on already strained modern power systems. To meet this rising load sustainably, regions such as New England can integrate Distributed Energy Resources (DERs) as localized generation. In this project, our goal is to determine effective DER placements within the IEEE 39-Bus System and develop dispatch strategies that minimize total operating cost while ensuring no transmission line exceeds 80% of its thermal capacity. Using a DC Optimal Power Flow (DC-OPF) framework, we demonstrate how DERs can reduce system costs and alleviate congestion, ultimately supporting a more efficient and reliable grid.

Problem Statement

As New England increases its reliance on renewable DERS, the transmission network faces new operational challenges. DERs can reduce dependence on traditional generators and lower system costs, but their decentralized injections can also alter power flows and potentially worsen congestion if not properly coordinated. In the IEEE 39-Bus system, this issue is magnified by a geographic imbalance: most generation is located in the northern portion of the network, while most loads lie in the south, creating heavily stressed north-to-south transmission paths. To maintain system reliability, the grid must ensure that no transmission line exceeds 80% of its thermal capacity, the threshold adopted for this project.

Methods and Analysis

1. Mapping

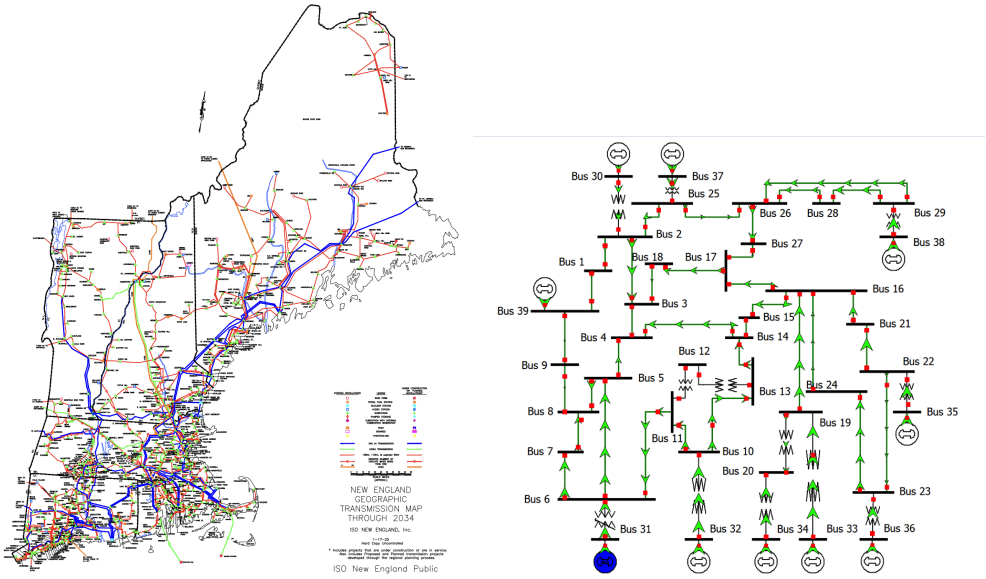


Figure 1: ISO New England Transmission Line Diagram (left), a 39-bus system diagram (right)



Table 6 Load and generation data for 39-bus system

Bus number	Load (MW)	Generation (MW)	Bus number	Load (MW)	Generation (MW)
1	97.6	0	21	274	0
2	0	0	22	0	0
3	322	0	23	247.5	0
4	500	0	24	308.6	0
5	0	0	25	224	0
6	0	0	26	139	0
7	233.8	0	27	281	0
8	522	0	28	206	0
9	6.5	0	29	283.5	0
10	0	0	30	0	1040
11	0	0	31	9.2	646
12	8.53	0	32	0	725
13	0	0	33	0	652
14	0	0	34	0	508
15	320	0	35	0	687
16	329	0	36	0	580
17	0	0	37	0	564
18	158	0	38	0	865
19	0	0	39	1104	1100
20	680	0			

Figure 2: ISO New England real map (left). Load and generation data of the 39 bus system(right).

To model and analyze the integration of DERs such as wind and solar farms into a large transmission system, we utilized the IEEE 39-bus test system. This system is a simplified representation of the New England power grid and provides a clear node-to-node structure that is easier to analyze than the full ISO-NE transmission diagram.

We began by reviewing line and bus data from Reference [2] together with the ISO-NE system map to identify where major generation and load centers are located. From the dataset, it is evident that most high-capacity generators are positioned in the upper portion of the network, while the largest loads are concentrated in the lower half. As a result, the tie-lines connecting these two regions must carry significant power transfers to meet demand. This observation guided our assumption that these lines are likely to operate near their limits. Therefore, we can assume that the intersection lines between the top and bottom sections of the 39 bus system will be working at capacity to satisfy the load and generation constraint in our DC-OPF program, as shown in the following:

$$\textbf{St: } P_{Gi} - P_{Li} = B\theta \qquad \textbf{i = 1, 2, 3, …N (positive integers)}$$

With P_{Gi} being the power generated by generator *i*, and P_{Li} being the load at node *i*. Furthermore, to meet the project goal of eliminating congestion in any lines in the network, we will be adding DERs at the load where it reaches the threshold value set for our project, which is anything greater than or equal to a load of 500 MW. This means that, depending on how close the actual load value compares to the threshold load, more than one DER may be added depending on the maximum power the system can generate. Based on the assumptions made in similar studies, i.e. reference [2], the project will be adding DERs into the system with no cost. This means that in our DC-OPF constraint, we will have the following for the cost component:

$$\textbf{min} \sum_{i=1}^{10} P_{Gi} C_{Gi} \qquad \textbf{i = 1, 2, 3, …N (positive integers)}$$

This is a key constraint for the DC-OPF program. We see that in order to obtain the most optimal power system, we shall always consider minimizing the cost of each generator. Meaning the order of supply by each generator will be greatly impacted by this constraint.

2. DC OPF Formulation:

To analyze power flows and generator dispatch, a DC-OPF model was created and solved with MATLAB's *linprog()* function. References [2] and [3] provide per-unit line parameters, allowing for direct incorporation of each value without recalculating them. This optimization will give us the total power being generated in correlation with the cost. Using the DC-OPF formulas, we obtain the following MATLAB inputs:

$$\min \sum_{i=1}^{10} C_{Gi} P_{Gi}$$

$$\text{s.to } P_{i,min} \leq P_{Gi} \leq P_{i,max} \qquad i = 1,2,3,..., N \text{ (positive integers)}$$

$$\sum_{i=1}^{10} P_{G,i} = P_{load,i}$$

$$-f_{max} \leq S (P_{G\,i} - P_{Li}) \leq f_{max}$$

This DC-OPF formulation represents the given original network, which produces the optimal network solution. This includes the total price and power generated from each generator to support the load requirement while simultaneously meeting the other constraints, such as power flow through lines. In this DC-OPF program, C_{Gi} represents the cost of each generator, P_{Gi} is the total power being produced by the generator, f_{max} is the line capacity, and S *is* the Power Transfer Distribution Factor (PTDF) matrix that allows us to calculate the line constraints without the need to include the angle at each generator. The S matrix can be implemented as the following with our model. Assumptions for picking the slack bus, which is the reference generator, have been taken to be the first generator:

$$S = [\, 0, D_X^{-1} A_r B_r^{-1} \,]$$

In the notation above, everything is given in matrix form, where S is an *N x N* matrix, D_X^{-1} is the inverse diagonal reactance matrix, A_r is the reduced incidence matrix, and B_r^{-1} is the reverse induced susceptance matrix. Note that A_r and D_x both follow the same sequence of the direction flow of each line.These directions have been mapped in accordance with the “from bus” and “to bus” data from reference [2].

3. Line Congestion Report Algorithm:

The DC-OPF alone is not sufficient to determine the best locations for DERs, so a congestion-detection algorithm was created and added to evaluate which lines were operating near their limits. Any line carrying 90% or more of its rated capacity was flagged as by the algorithm. We selected 90% as the threshold to ensure the system retains enough margin to handle unexpected fluctuations in load, and identify lines that are not only under complete congestion, but also approaching complete congestion. This algorithm generates a visual comparison of actual versus maximum line loading, which helps identify where new DER capacity would most effectively relieve stress on the network, and its formulation is the following:

```
abs_flow   = abs(f_line)
loading    = abs_flow ./ fmax
congested  = find(abs_flow >= fmax - tol)
heavily    = find(loading >= 0.9)
```

As we can see from the algorithm above, the line congestion parameter is set to 90%, which means that any lines reaching 90% of their maximum capacity will be in the zone of congestion. Our choice of 90% is considered as congestion to provide room for the network system to handle any unexpected loads. With the given code implementation, we will then be able to displace a bar plot showing the maximum capacity of each line and the capacity the line is transmitting at. This result and plot will be demonstrated in the simulation of the report.

4. Conducting DC OPF for various approaches to obtain zero line congestions.

In order to successfully obtain a balanced 39 bus system network with no line congestion and optimal cost, we conducted an analysis on the displacement of DER systems. In this project, we chose to use wind farms and solar farms as our choices of DER systems, as referred to in references [4] and [5], because they are popularly used across the country. Recall that, when adding DERs into a network, assumptions can be made that their costs are free to use. This implies that, with the initial set of constraints, nothing must be altered or added to obtain the correct DC-OPF

algorithm. In real-world applications, there are many costs associated with building a DER at given locations; however in the case of this study and in accordance with assumptions made in reference [2], the DERs are free to use. The actual values of our cost will, in turn, change according to the DERs we add into the system. Adding the DERs will reduce the power being generated by the generator, since the DERs are free to use, and the 39 bus system loads will try to consume all the power being generated by the lower-cost DERs. The bulk of the cost constraints do not need to be changed, as the following shows:

$$\min \sum_{i=1}^{10} P_{Gi} C_{Gi} \qquad i = 1, 2, 3, \ldots N \text{ (positive integers)}$$

However, the additions of DERs will impact other constraints in the DC-OPF model, such as the line congestion constraints and the load requirement constraints. Since we are analyzing the system as a whole, we do not need to consider net power changes at each node; only the sum with respect to the whole system. The changes to the DC-OPF program are shown as follows.

$$\min \sum_{i=1}^{10} C_{Gi} P_{Gi}$$

$$\text{s.to } P_{i,min} \leq P_{Gi} \leq P_{i,max} \qquad i = 1,2,3,\ldots, N \text{ (positive integers)}$$

$$\sum_{i=1}^{10} P_{G,i} = P_{load,i} - P_{renew\ i}$$

$$-f_{max} \leq S\,((P_{G\,i} + P_{renew\ i}) - P_{Li}) \leq f_{max}$$

From the above, we observe the only changes that need to be added to the DC OPF program after adding the DERs at the load. With the loads hitting the threshold, the ideal power that a DER system can produce under the conditions of the model will be subtracted from the load. Another method with a similar outcome is to add P_{rewew} to the respective generated powers.

4A. (Approach 1): Adding DERs at the two ends of the congested lines:

The initial strategy was to place DERs directly at the buses connected by a congested line; in this case, buses 2 and 3 were the sole affected buses. While this reduced loading on that particular line, it unexpectedly caused congestion elsewhere in the network. Since DERs are assumed free to operate, the lower-cost power they generate is heavily utilized, causing large transfers toward the load-heavy lower half of the system. This increases the burden on other transmission paths and ultimately worsens overall congestion. Although this approach reduces total system cost, it does not resolve the congestion issue, so it was not considered viable.

4B. (Approach 2): Adding DERs at the load that's meeting our threshold value of 500MW or greater:

We next adopted a strategy commonly seen in real system planning: placing generation near major loads to prevent excessive reliance on distant generators and long transmission paths. Inspired by how data centers are often built alongside new substations to avoid stressing existing infrastructure, we applied the same logic to the 39-bus system. With this motivation, the same method was applied in our project. With the load threshold value of 500 MW, we located all the nodes that have a load of 500MW or greater and have a prototype of adding an existing DER system at that node.

Project / Farm	Location / State
Three Corners Solar	Kennebec County, Maine (towns Benton, Clinton, Unity)
Nautilus Solar Sanford Community Solar	Sanford, Maine
Happy Hollow Community Solar + Storage Farm	Winchendon, Massachusetts
Ware-Palmer Solar Farm	Ware, Massachusetts
Monson Solar (Westview Farm)	Monson / Westview Farm, Massachusetts
Somers Solar Center	Somers, Connecticut

Project	Approx. capacity (MW)
Vineyard Wind 1	~806 MW
Revolution Wind	~704 MW
SouthCoast Wind (project / selection)	~1,287 MW (project total; MA selected ~1,087 MW of that)
New England Wind 1	~791 MW

Figure 4: Summary of known solar farms in the New England region(Left). Summary of Wind farms in the New England region(Right).

For each bus with a load of 500 MW or more, we added a DER sized accordingly to realistic wind and solar farm capacities listed in references [4] and [5]. In some cases, a combination of both technologies was used to match larger loads. These DER outputs were subtracted from the corresponding loads in the DC-OPF model. This method produced the best results: the system achieved no line congestion, feasible power flows, satisfied load requirements, and a reduced overall operating cost due to the zero-cost DERs. Additionally, most lines operated below 90% loading, leaving sufficient margin for variations or unexpected load increases.

Simulations and Results

The following are the simulated figures that represent the findings of both DER allocation simulations, as well as the conditions of the system under the initial DC OPF conditions:

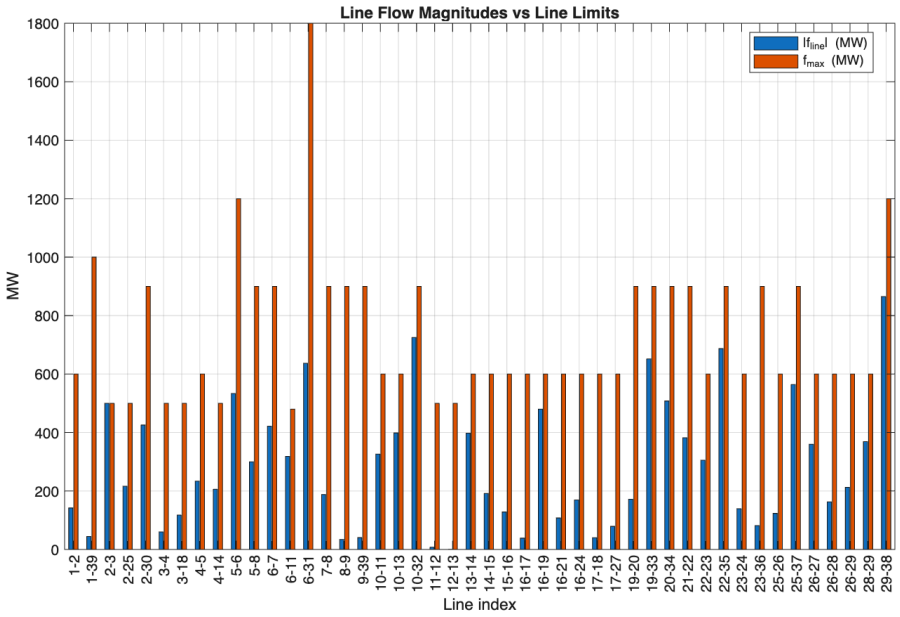


Figure 5.1 - Line Connection(x-axis) vs line flow(blue) and line limits(orange) (y-axis) in MW. This is for our 39-bus system with no DER's added. Line 2-3 is at limit, 10-12 & 16-19 is above 80% its limit.

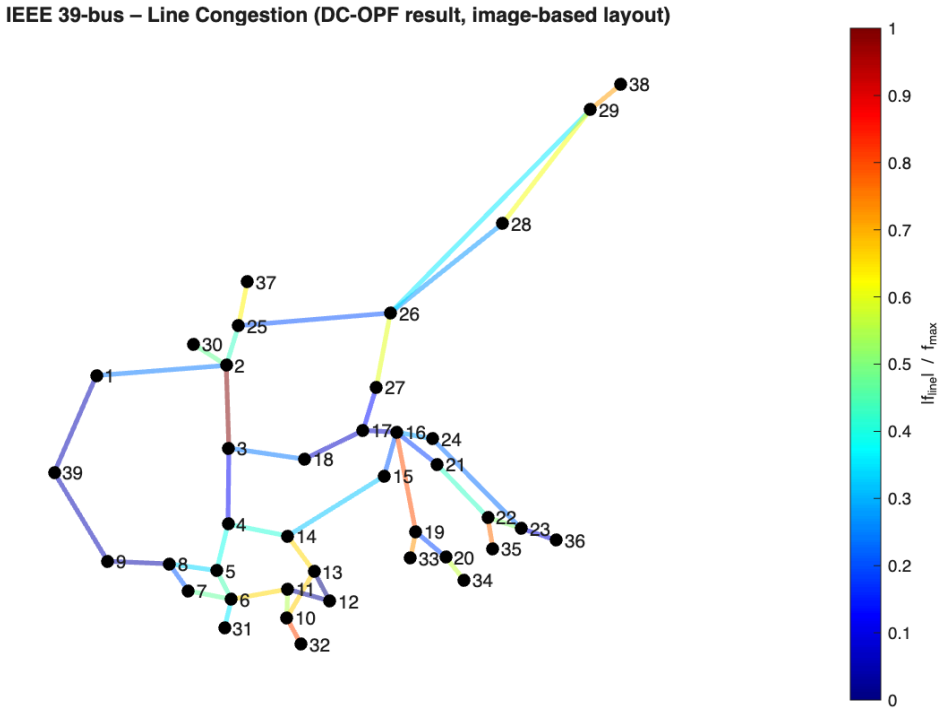


Figure 5.2 - System mapping view of our original 39-bus system. Red indicates congestion, blue indicates low line usage.

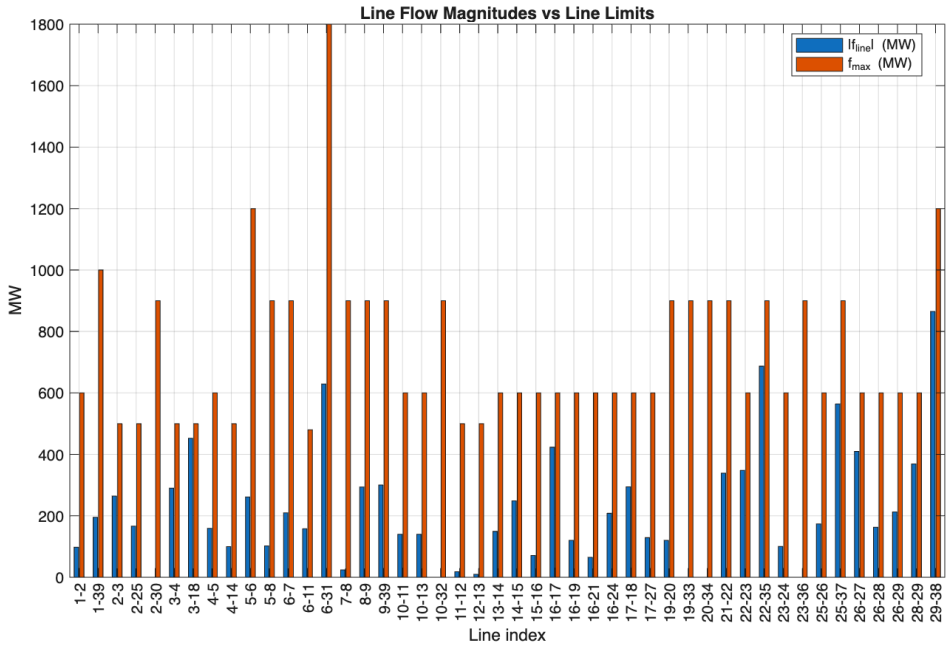


Figure 6.1 - Line Connection(x-axis) vs line flow(blue) and line limits(orange) (y-axis) in MW. This is for our 39-bus system with 6 DER's added. Only line 2-3 is above 80% capacity(90.47%).

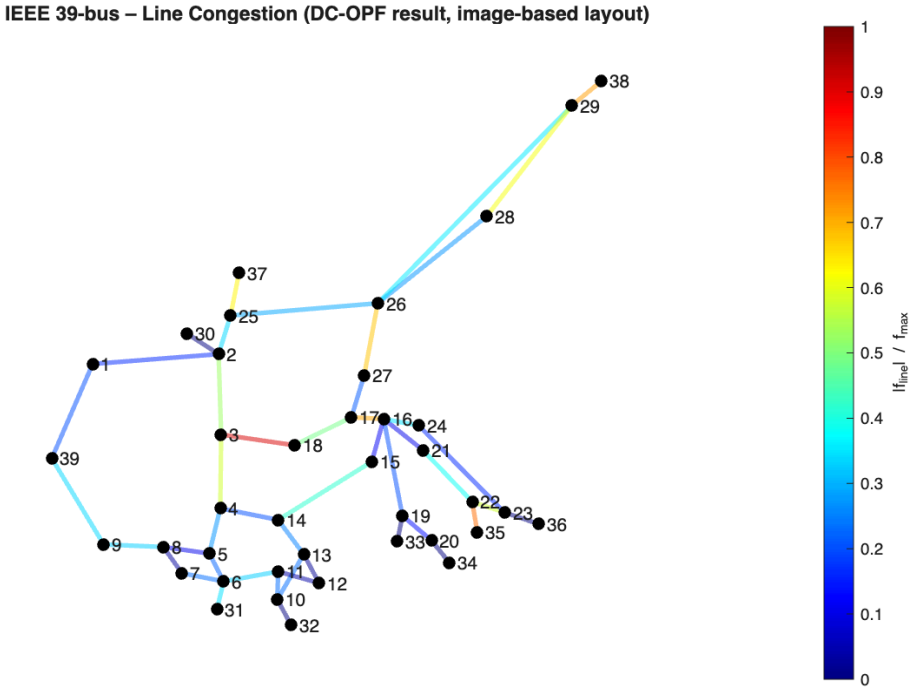


Figure 6.2 - System mapping view of our 39-bus system with the added DER's. Red indicates congestion, blue indicates low line usage.

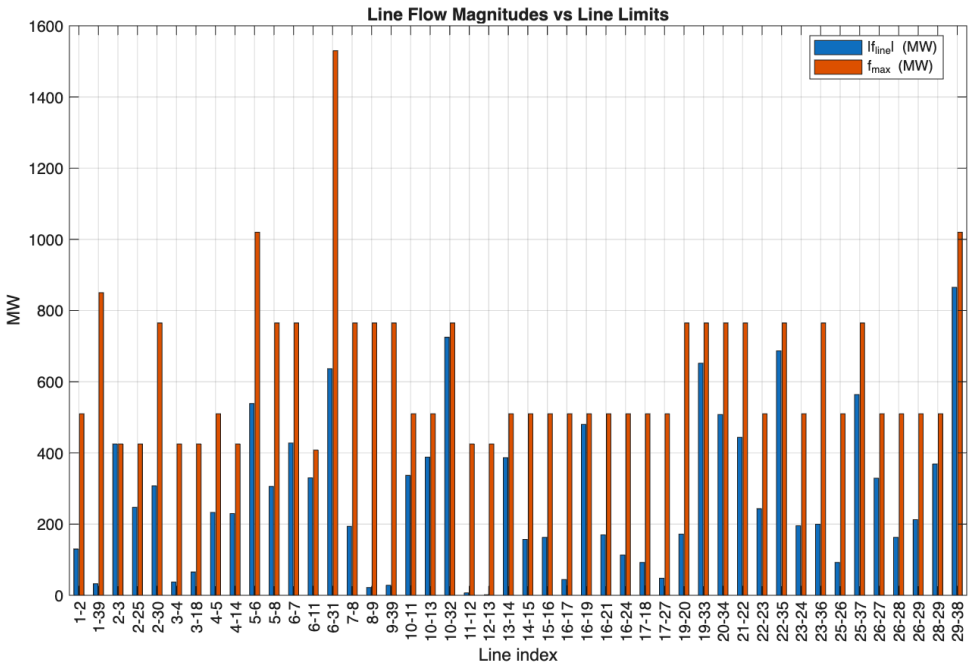


Figure 7.1 - Line Connection(x-axis) vs line flow(blue) and line limits(orange) (y-axis) in MW. This is for our 39-bus system with no DER's added and lines limited to 85% their typical transmission limit. Line 2-3 is at capacity, and 6 other lines are above 80% capacity.

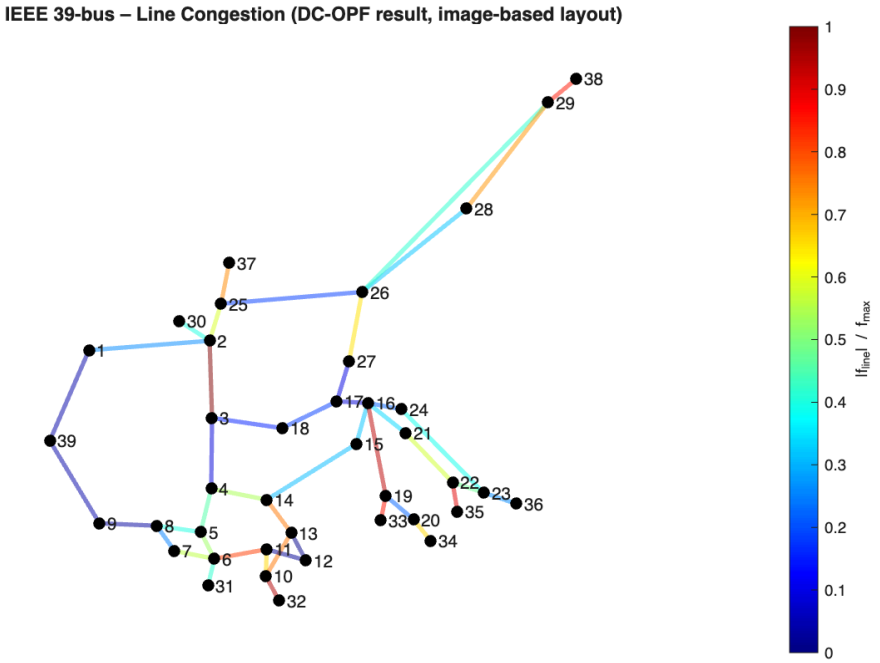


Figure 7.2 - System mapping view of our congested 39-bus system with no DER's. Red indicates congestion, blue indicates low line usage.

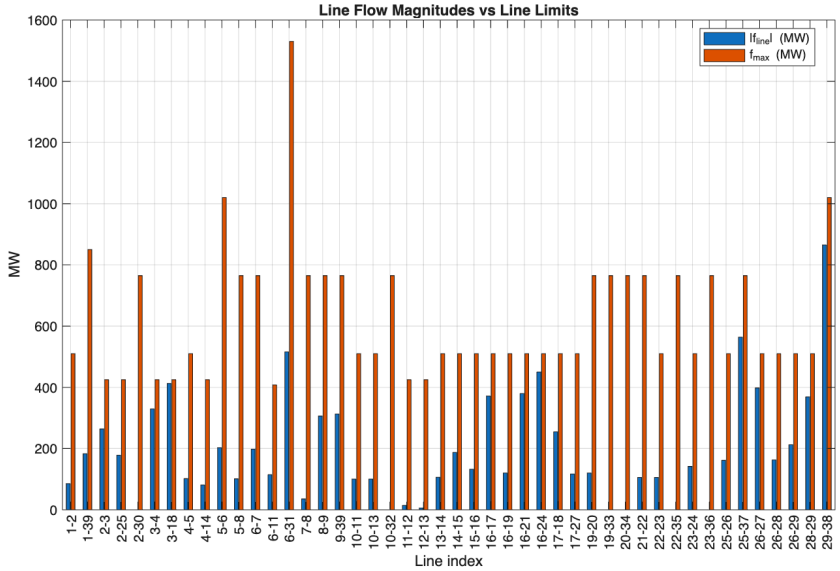


Figure 8.1 - Line Connection(x-axis) vs line flow(blue) and line limits(orange) (y-axis) in MW. This is for our 39-bus system with 7 DER's added and lines limited to 85% their typical transmission limit. Only 3 lines are above 80% capacity.

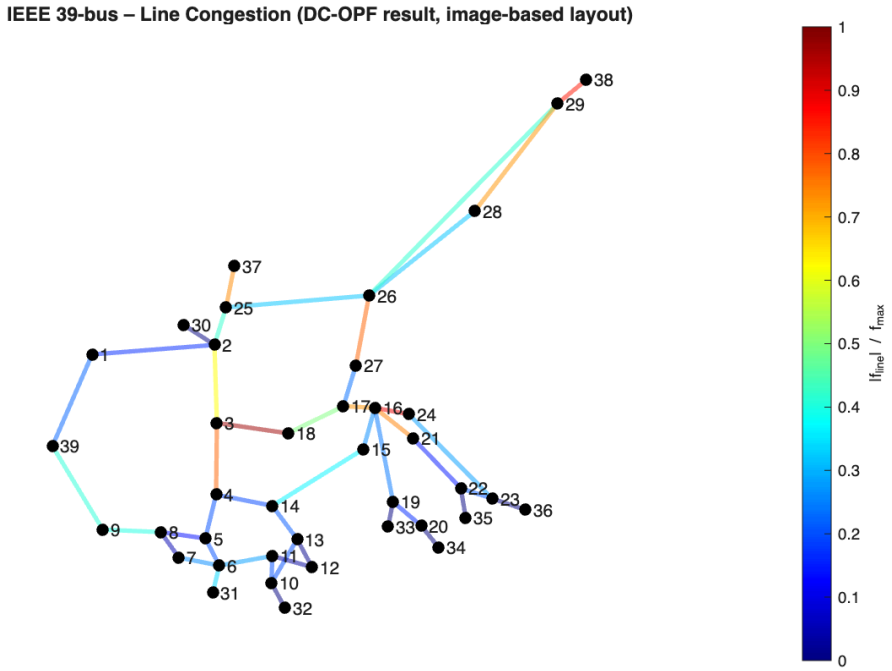


Figure 8.2 - System mapping view of our congested 39-bus system with no DER's. Red indicates congestion, blue indicates low line usage.

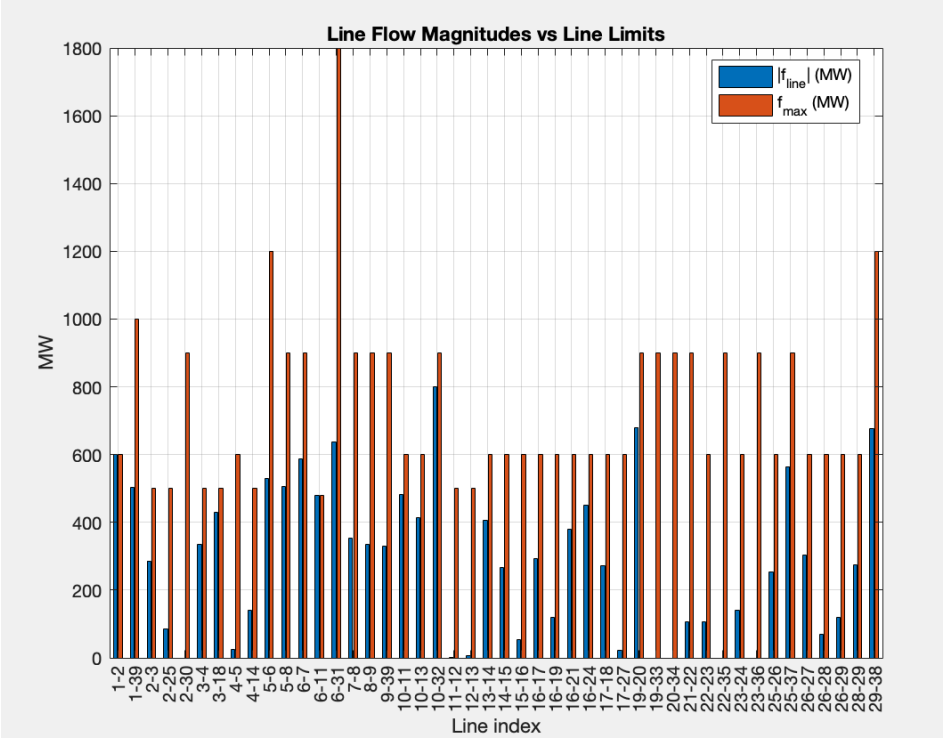


Figure 9.1 - Line Connection(x-axis) vs line flow(blue) and line limits(orange) (y-axis) in MW. This is for our 39-bus system with DER's added at the two nodes connected congestion line 2 to 3.

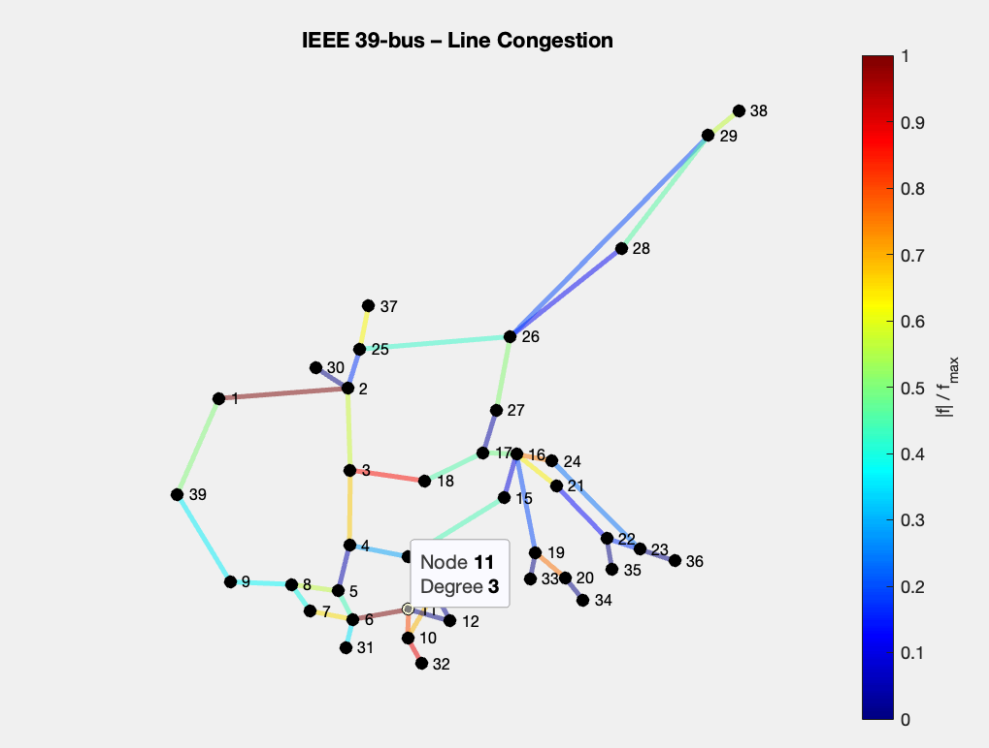


Figure 9.2 - System Mapping view of our congested 39- bus system with DER's added at the congested line nodes. Red indicates congestion, blue indicates low line usage.

Tasks and Deliverables

The following is a list of the tasks that were outlined and completed for the project. All tasks were met, implemented, and/or simulated to the level introduced in the project's scope:

1. Formulate DC-OPF model for New England IEEE 39-bus system incorporating generator costs, line flow constraints, and bus load data. **(100%)**
2. Set simulation thresholds for ideal line congestion (≤ 85%) and load size (≥ 500 MW). **(100%)**
3. Calculate line congestions for baseline 39-bus system by running DC-OPF model and utilizing PTDF matrix to calculate real power flows across transmission lines.**(100%)**
4. Compare power flow magnitudes to max power flow ratings interpolated from data to identify lines under or exceeding complete congestion (≥ 100%). **(100%)**
5. Re-run DC-OPF using congested-line-connected bus implementation, placing DER's at identified buses connected by lines under congestion and evaluating changes in power flows. **(100%)**
6. Re-run DC-OPF using highest-load-capacity bus implementation, placing DER's at identified buses that account for the largest loads in the system- exceeding the set size threshold (500 MW)- and evaluating changes in power flows. **(100%)**
7. Compare effectiveness of both DER allocation methodologies: Analysis implies direct net load reduction at larger buses is a more viable solution than attempting to directly alleviate congested lines at their respective buses.**(100%)**
8. Run DC-OPF with adjusted constraints for ideal line congestion (≤ 85%) instead of complete (≥ 100%) to analyze where additional DER's should be allocated **(100%)**
9. Map line congestion to geographical locations, using bus X-Y data from IEEE 39-bus system image, for congestion visualization. **(100%)**

Conclusions

This project examined how Distributed Energy Resources (DERs) can be added to the IEEE New England 39-bus system to reduce transmission congestion and lower the overall operating cost. Using the DC-OPF model we constructed, we first studied the original network and found several lines operating close to their limits. The most congested line was between buses 2 and 3, and other lines such as 10 to 12 and 16 to 19 also exceeded 80 percent of their maximum capacity. These results showed that the system experiences significant stress under normal loading conditions.

To address this problem, we tested two main DER placement strategies. The first strategy added DERs at the buses that were directly connected to the congested lines. Again, although this reduced loading on those specific lines, it caused new congestion to appear in other parts of the network. Because of this, the first strategy was not effective for improving the system as a whole. The second strategy added DERs at the buses with the largest loads, which in this project were defined as loads of 500 megawatts or more. This method reflected how real power systems often place new generation near large loads. When DERs were added to these high-load buses, the results improved significantly. Line loadings decreased throughout the network, no lines exceeded the 90 percent congestion threshold, and the total cost of supplying the load decreased because the DERs were modeled as zero-cost resources.

The simulations showed that placing DERs based on load size created a more balanced flow of power across the north and south sections of the system. The line flow graphs and the congestion maps also confirmed that this strategy provided more consistent relief than placing DERs near the originally congested lines.

In summary, the project demonstrates that it is more effective to place DERs at large loads rather than at congested lines when trying to reduce congestion and minimize cost. If this project were continued, future work could include adding realistic costs for DER construction and operation, studying how renewable output changes over time, and testing a full AC power flow model to capture more detailed system behavior.

References

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